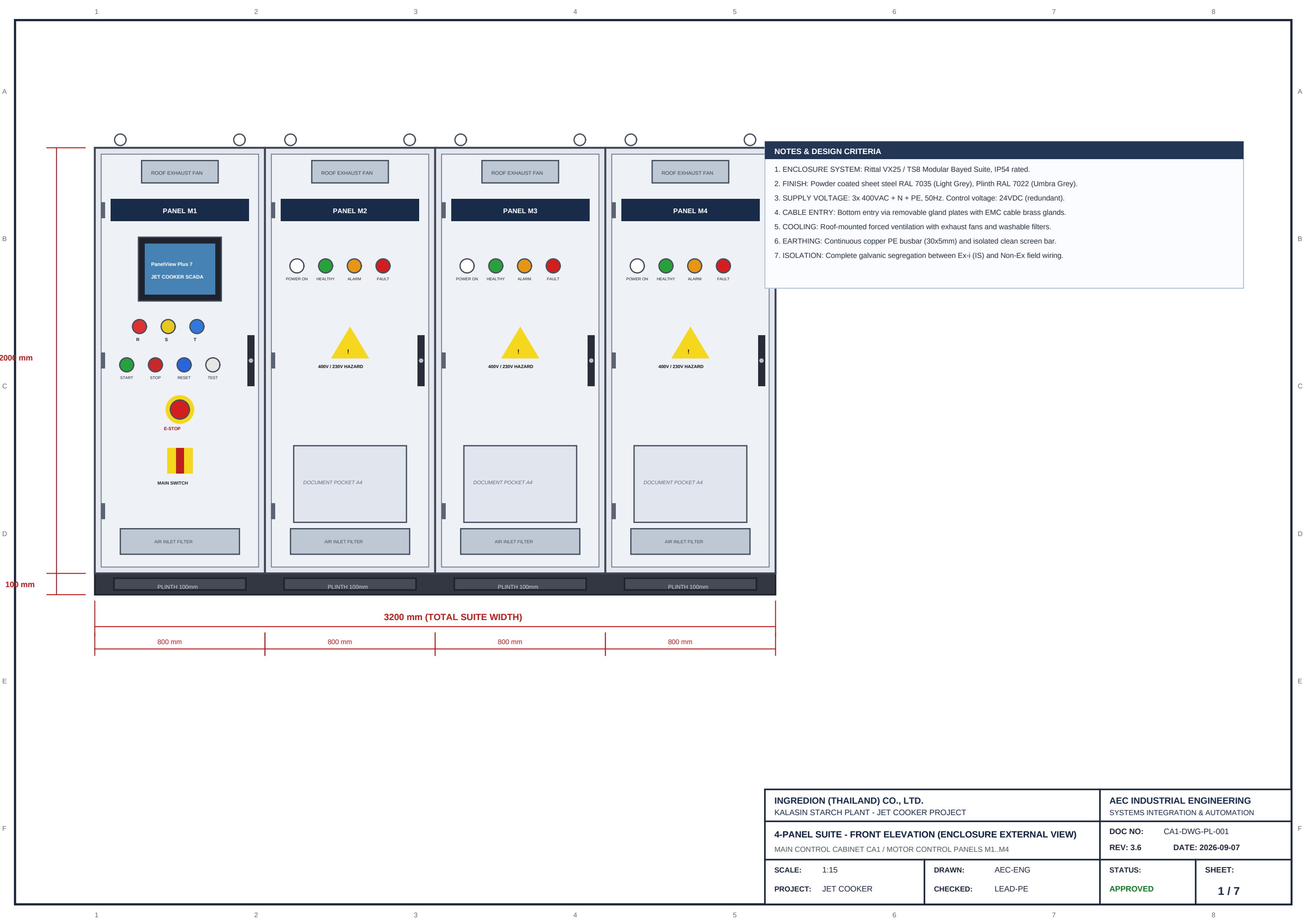




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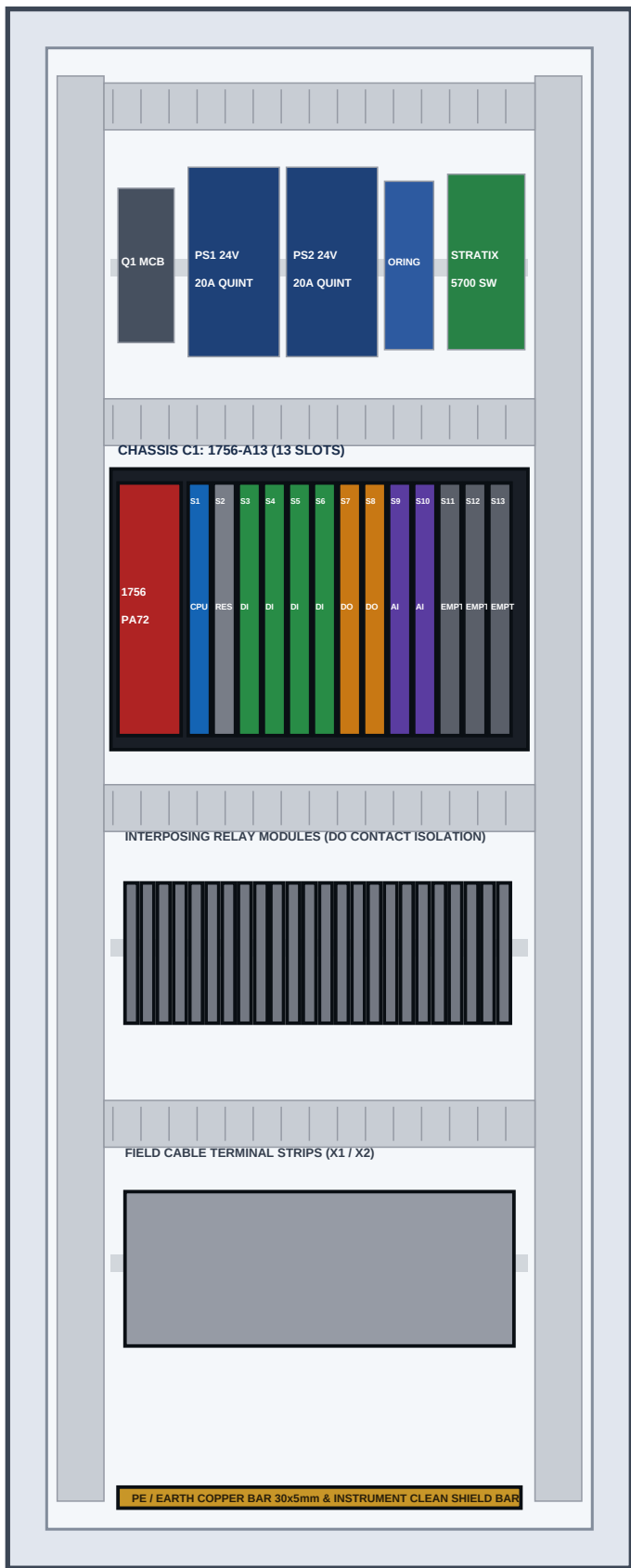
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NOTES & DESIGN CRITERIA			
1. CHASSIS SYSTEM: Rockwell Automation Allen-Bradley ControlLogix 1756-A13 (13-slot racks).			
2. WIRING SEGREGATION: 50mm min. clearance between 230VAC, 24VDC, and analog signal paths.			
3. INTRINSIC SAFETY: Panels M3 & M4 feature dedicated blue IS terminal strips and 50mm physical air gap partition plates complying with IEC 60079-11 / IEC 60079-14 Ex-i standards.			
4. SPARE CAPACITY: All 4 panels maintain >25% spare DIN rail space and spare card slots.			
5. WIRE DUCT: Halogen-free slotted PVC wire duct with snap-on covers (60x80mm & 80x100mm).			

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT			AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
4-PANEL SUITE - INTERNAL GENERAL ARRANGEMENT (DOORS OPEN) MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4			DOC NO: CA1-DWG-PL-002 REV: 3.6	DATE: 2026-09-07
SCALE: 1:15 PROJECT: JET COOKER	DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED		SHEET: 2 / 7



Panel M1 - Rack C1 Hardware Allocation & Field Terminal Schedule							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-L950TPSXT	ControlLogix 5580 Controller (50MB Memory)	CPU	1	ETH1/2	Plant Master Controller / SCADA	ACTIVE (MASTER)
02	Reserve	Reserved for 1756-EN2TR / Redundancy	RES	-	-	Reserved for Controller Redundancy	SPARE (RESERVE)
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S03-X1,X2	JB-401 Field Signals & Spares	ACTIVE
04	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S04-X1,X2	JB-401 Field Signals	ACTIVE
05	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S05-X1,X2	JB-601 Field Signals	ACTIVE
06	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C1S06-X1,X2	JB-601 Field Signals & CA1 Spares	ACTIVE
07	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C1S07-X1,X2	JB-401 Solenoid Valves & DO	ACTIVE
08	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C1S08-X1,X2	JB-601 Solenoids & CA1 Spares	ACTIVE
09	1756-IF16	16-Point High Resolution Analog Input	AI	16	C1S09-X1	JB-401 Analog Transmitters (4-20mA)	ACTIVE
10	1756-IF16	16-Point High Resolution Analog Input	AI	16	C1S10-X1	JB-601 Analog Transmitters (4-20mA)	ACTIVE
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)

PANEL M1 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	- FIELD DESTINATIONS: Connects to field junction boxes: JB-401, JB-601, CA1.
	- POWER INFEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.
	- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.
	- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).
	- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT		AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M1 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1..M4		DOC NO: CA1-DWG-PL-003 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8 PROJECT: JET COOKER	DRAWN: AEC-ENG CHECKED: LEAD-PE	STATUS: APPROVED	SHEET: 3 / 7

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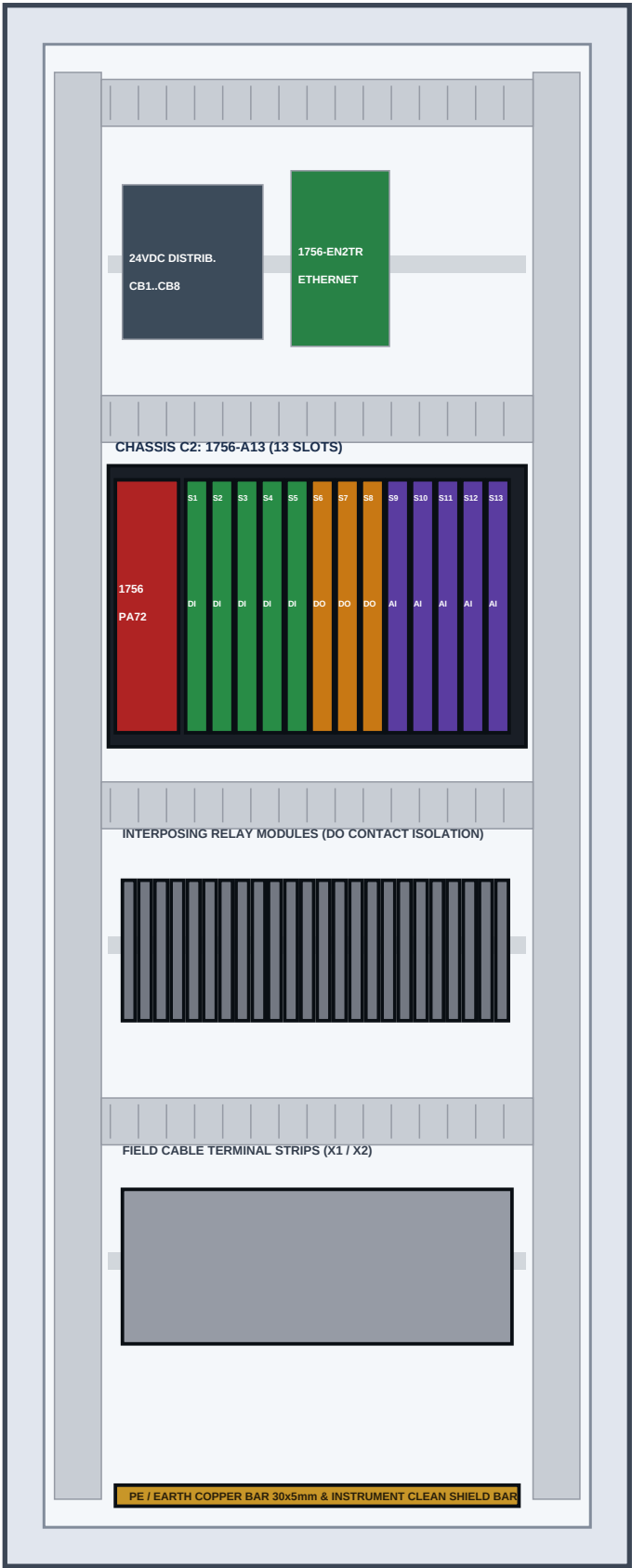
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PANEL M2 - RACK C2 HARDWARE ALLOCATION & FIELD TERMINAL SCHEDULE							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S01-X1,X2	JB-402 & JB-602 Flow & Limit Switches	ACTIVE
02	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S02-X1,X2	JB-602 & JB-607 Field Inputs	ACTIVE
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S03-X1,X2	JB-607 Spray Dryer 7th Flr DI	ACTIVE
04	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S04-X1,X2	JB-607 Spray Dryer 7th Flr DI	ACTIVE
05	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C2S05-X1,X2	JB-607 Spray Dryer 7th Flr DI	ACTIVE
06	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C2S06-X1,X2	JB-402 & JB-602 On/Off Valves	ACTIVE
07	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C2S07-X1,X2	JB-607 Solenoid Valves (DO)	ACTIVE
08	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C2S08-X1,X2	JB-607 Solenoid Valves (DO)	ACTIVE
09	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S09-X1	JB-402 Jet Cooker Flow/Press AI	ACTIVE
10	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S10-X1	JB-602 Spray Dryer 3rd Flr AI	ACTIVE
11	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S11-X1	JB-602 Spray Dryer 3rd Flr AI	ACTIVE
12	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S12-X1	JB-607 Spray Dryer 7th Flr AI	ACTIVE
13	1756-IF16	16-Point High Resolution Analog Input	AI	16	C2S13-X1	JB-607 Spray Dryer 7th Flr AI	ACTIVE

PANEL M2 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door. - FIELD DESTINATIONS: Connects to field junction boxes: JB-602, JB-402, JB-607. - POWER INFEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus. - NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection. - CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP). - SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M2 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-004 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8	DRAWN: AEC-ENG		STATUS: APPROVED		SHEET: 4 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE				

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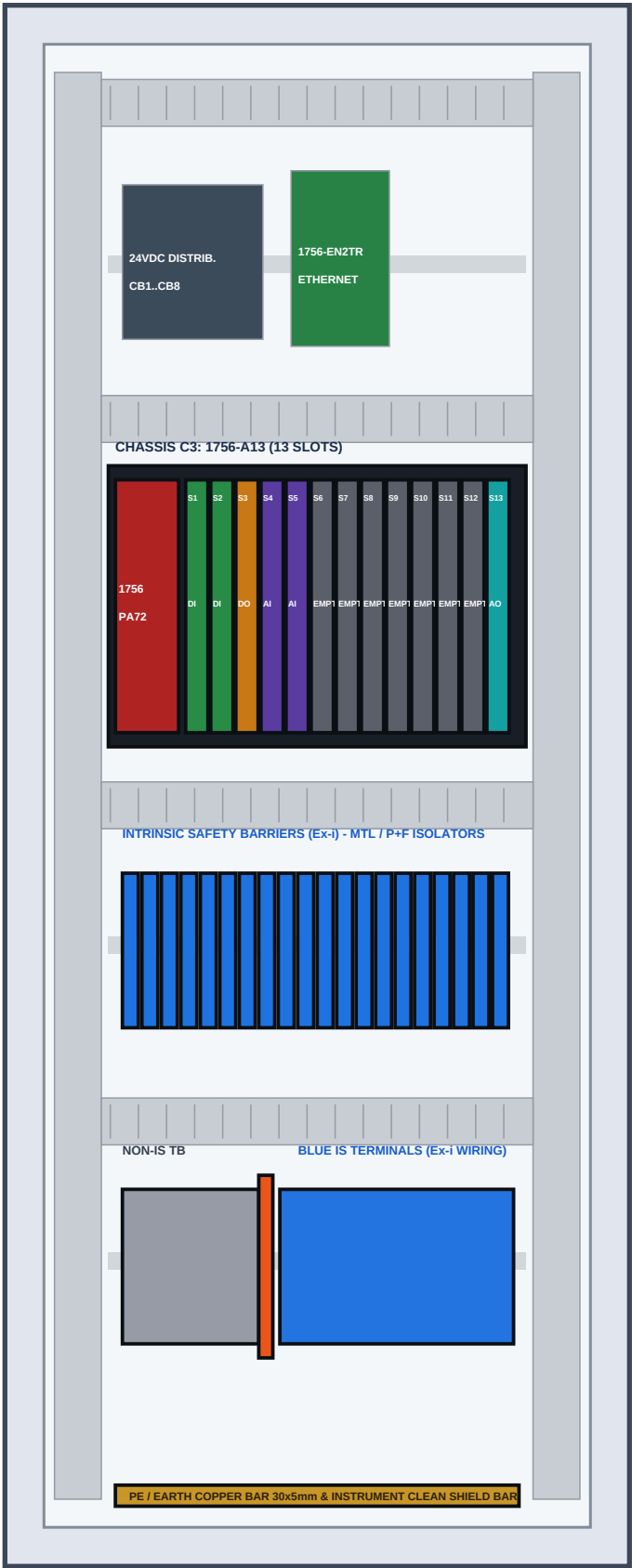
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PANEL M3 - RACK C3 HARDWARE ALLOCATION & FIELD TERMINAL SCHEDULE							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C3S01-X1,X2	IS-JB-603 & IS-JB-618 Ex-i DI	ACTIVE (IS)
02	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C3S02-X1,X2	IS-JB-618 & JB-618 Field DI	ACTIVE
03	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C3S03-X1,X2	JB-618 Packing Tower 8th Flr DO	ACTIVE
04	1756-IF16	16-Point High Resolution Analog Input	AI	16	C3S04-X1	IS-JB-603 Ex-i Transmitters (AI)	ACTIVE (IS)
05	1756-IF16	16-Point High Resolution Analog Input	AI	16	C3S05-X1	IS-JB-618 & JB-618 Transmitters	ACTIVE (IS)
06	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
07	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
08	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
09	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
10	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-OF8	8-Point High Resolution Analog Output	AO	8	C3S13-X1	Control Valves / Speed References	ACTIVE

PANEL M3 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	
- FIELD DESTINATIONS: Connects to field junction boxes: IS-JB-603, IS-JB-618, JB-618.	
- POWER INFEEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.	
- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.	
- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).	
- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M3 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-005 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8	DRAWN: AEC-ENG		STATUS: APPROVED		SHEET: 5 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE				

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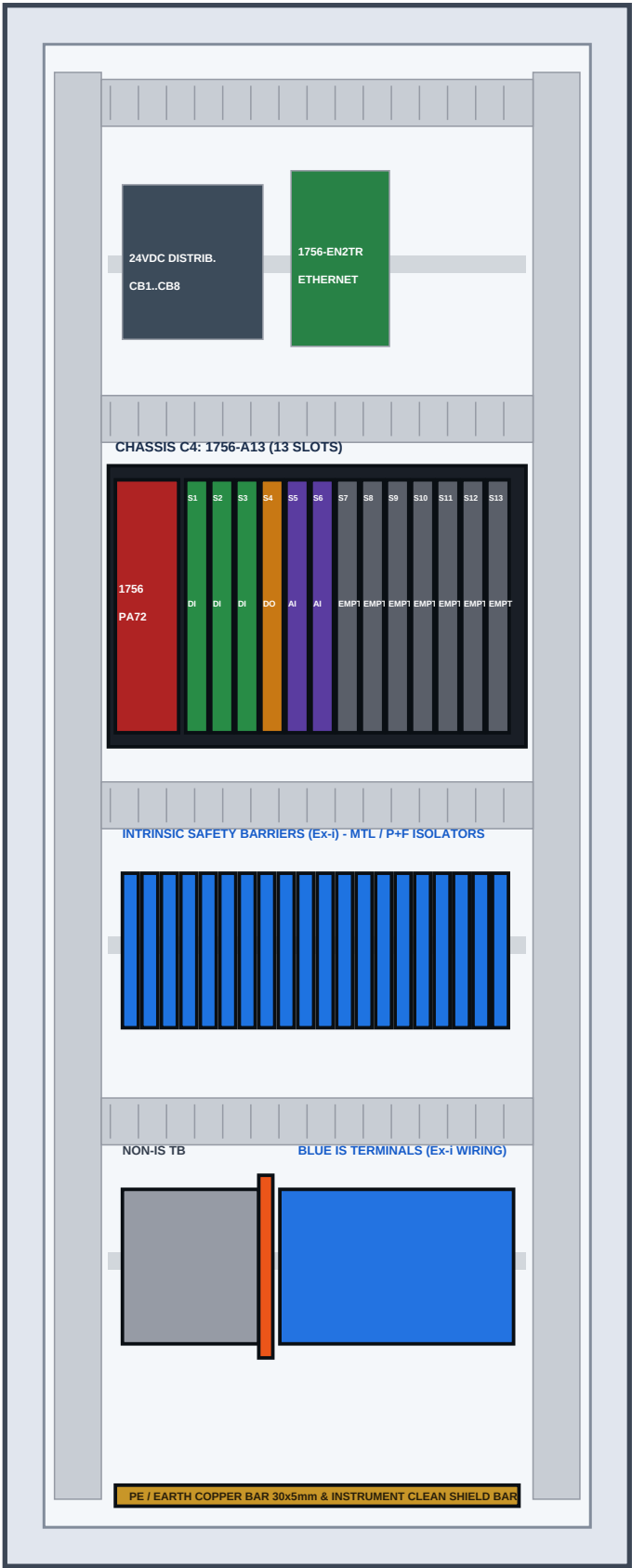
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PANEL M4 - RACK C4 HARDWARE ALLOCATION & FIELD TERMINAL SCHEDULE							
Slot	Module Catalog	Description / Module Function	I/O Type	Channels	Terminal Strip	Connected Junction Boxes / Signals	Status
PS	1756-PA72	ControlLogix 85-265V AC Power Supply (72W @ 60C)	POWER	120/230VAC	Internal Rack	Chassis Backplane Power	ACTIVE (RED)
01	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C4S01-X1,X2	JB-606 & IS-JB-612 Ex-i DI	ACTIVE
02	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C4S02-X1,X2	IS-JB-612 & JB-612 Field DI	ACTIVE (IS)
03	1756-IB32	32-Point 24VDC Sink/Source Digital Input	DI	32	C4S03-X1,X2	IS-JB-608 & JB-608 Ex-i DI	ACTIVE (IS)
04	1756-OB32	32-Point 24VDC Source Digital Output	DO	32	C4S04-X1,X2	JB-606, JB-612, IS-JB-608 DO	ACTIVE
05	1756-IF16	16-Point High Resolution Analog Input	AI	16	C4S05-X1	JB-606 & IS-JB-612 Ex-i AI	ACTIVE (IS)
06	1756-IF16	16-Point High Resolution Analog Input	AI	16	C4S06-X1	JB-612, IS-JB-608, JB-608 AI	ACTIVE (IS)
07	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
08	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
09	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
10	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
11	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
12	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)
13	1756-N2	Slot Filler Plate (Empty Slot)	EMPTY	-	-	Chassis Spare Expansion Slot	SPARE (AVAILABLE)

PANEL M4 INSTALLATION SPECIFICATIONS & FIELD CONNECTIONS	
- ENCLOSURE SIZE: 800 mm (W) x 2000 mm (H) x 800 mm (D) + 100 mm plinth base. Single door.	
- FIELD DESTINATIONS: Connects to field junction boxes: JB-606, IS-JB-612, JB-612, IS-JB-608, JB-608.	
- POWER INFEEED: Dual redundant 24VDC feeder lines from Panel M1 distribution bus.	
- NETWORK ARCHITECTURE: Dual-ring EtherNet/IP Device Level Ring (DLR) connection.	
- CABLING RULES: All analog inputs wired with individually shielded twisted pairs (LiY-CY TP).	
- SHIELD GROUNDING: Cable screens grounded at panel entry copper earth bar only (single-point earthing).	

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
PANEL M4 - SUBPANEL MOUNTING PLAN & COMPONENT SCHEDULE MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-006 REV: 3.6	DATE: 2026-09-07
SCALE: 1:8	DRAWN: AEC-ENG		STATUS: APPROVED		SHEET: 6 / 7
PROJECT: JET COOKER	CHECKED: LEAD-PE				

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MAJOR HARDWARE BILL OF MATERIALS (BOM) - ENCLOSURES, PLC HARDWARE & ACCESSORIES								
Item	Part Number / Catalog	Manufacturer	Component Description	Installed Location	Qty	Spare Qty	Technical Specifications / Ratings	Remarks
1	VX 8806.500	Rittal	Industrial Bayed Enclosure Suite (4 bays)	M1, M2, M3, M4	4	-	800x2000x800 mm, IP54, Sheet Steel RAL 7035	Complete with side panels &
2	1756-A13	Rockwell Automation	ControlLogix 13-Slot Chassis	M1, M2, M3, M4	4	-	Standard backplane, horizontal mounting	Chassis C1, C2, C3, C4
3	1756-PA72	Rockwell Automation	ControlLogix Power Supply	M1, M2, M3, M4	4	1	85-265V AC, 72W output @ 60 deg C	One supply per chassis
4	1756-L950TPSXT	Rockwell Automation	ControlLogix 5580 Controller	Panel M1 (C1S01)	1	1	Extreme environment, 50MB memory, 1Gbps ETH	Master Plant Controller
5	1756-EN2TR	Rockwell Automation	EtherNet/IP Dual-Port DLR Comm Module	M2, M3, M4	3	1	10/100 Mbps, 128 TCP connections, DLR ring	I/O Remote Adapter modules
6	1756-IB32	Rockwell Automation	32-Point 24VDC Digital Input Module	M1, M2, M3, M4	14	2	24VDC Sink/Source, 10-31.2VDC, Isolated groups	544 total DI channels availa
7	1756-OB32	Rockwell Automation	32-Point 24VDC Digital Output Module	M1, M2, M3, M4	7	1	24VDC Source, 10-31.2VDC, 0.5A per point	224 total DO channels availa
8	1756-IF16	Rockwell Automation	16-Point Analog Input Module	M1, M2, M3, M4	11	1	Current / Voltage (4-20mA, 0-10V), 16-bit resolution	176 total AI channels availa
9	1756-OF8	Rockwell Automation	8-Point Analog Output Module	Panel M3 (C3S13)	1	1	Current / Voltage (4-20mA, 0-10V), 15-bit resolution	8 total AO channels availabl
10	1756-N2	Rockwell Automation	ControlLogix Slot Filler Plate	M1, M3, M4	17	-	Chassis slot cover plate for dust/cooling protection	Covers unpopulated slots
11	QUINT4-PS/1AC/24DC/20	Phoenix Contact	Primary Switched Redundant 24VDC Power Supply	Panel M1	2	1	Input 100-240VAC, Output 24VDC 20A, SFB Technology	Primary/Secondary Redundant
12	QUINT4-DIODE/40	Phoenix Contact	Diode Redundancy Module	Panel M1	1	-	Redundancy decoupling module 2x 20A / 1x 40A	Decouples PS1 & PS2
13	Stratix 5700	Rockwell Automation	Industrial Managed Ethernet Switch	Panel M1	1	-	16 Ports 10/100 Mbps, DLR, CIP Sync, VLAN support	Panel Network Backbone
14	KFD2-STC4-Ex1	Pepperl+Fuchs	Intrinsically Safe SMART Transmitter Isolator	Panel M3, M4	38	5	Ex-ia IIC, 4-20mA current repeater with HART support	For IS-JB-603, 608, 612, 618
15	KFD2-SL2-Ex2	Pepperl+Fuchs	Intrinsically Safe Digital Switch Isolator	Panel M3, M4	46	6	Ex-ia IIC, Dual-channel dry contact / NAMUR sensor	For IS Level & Limit Switche
16	Finder 38.51	Finder	Interposing Relay Module (DO Isolation)	M1, M2, M3, M4	120	15	24VDC Coil, 1 CO 6A contact, LED indicator	DIN Rail mount relay interfa
17	WDU 2.5 / WDU 4	Weidmuller	Feed-Through Terminal Blocks (Gray)	M1, M2, M3, M4	950	100	800V, 24A, Screw connection, multi-tier	Standard Field Wiring
18	WDU 2.5 BL	Weidmuller	Intrinsically Safe Terminal Blocks (Blue)	Panel M3, M4	320	50	Ex-i Blue colored terminal blocks for hazardous area	IS Field Wiring Only
19	SK 3328.540	Rittal	Roof Mounted Cooling Unit / Fans	M1, M2, M3, M4	4	-	230V, 50/60Hz, 1500W cooling capacity / high flow	Temperature maintenance < 35

INGREDION (THAILAND) CO., LTD. KALASIN STARCH PLANT - JET COOKER PROJECT				AEC INDUSTRIAL ENGINEERING SYSTEMS INTEGRATION & AUTOMATION	
4-PANEL SUITE - COMPLETE BILL OF MATERIALS (BOM) & SPECIFICATION MAIN CONTROL CABINET CA1 / MOTOR CONTROL PANELS M1...M4				DOC NO: CA1-DWG-PL-007 REV: 3.6	DATE: 2026-09-07
SCALE: N/A	DRAWN: AEC-ENG		STATUS:		SHEET:
PROJECT: JET COOKER	CHECKED: LEAD-PE		APPROVED		7 / 7

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